a Allocation contrasts across pools, budgets and models (mIoU points)

	frozen DINOv2 + MLP · balanced pool			frozen DINOv2 + MLP · skewed pool			complete model · skewed
	B = 6	B = 12	B = 24	B = 6	B = 12	B = 24	B = 12
GeoK − visual K	-0.24 [-0.6, +0.0]	**+0.90** [+0.4, +1.5]	+0.17 [-0.3, +0.6]	+0.17 [-0.3, +0.6]	**+1.33** [+0.7, +2.0]	+0.32 [-0.1, +0.8]	**+1.68** [+0.9, +2.5]
GeoK − placebo quota	-0.04 [-0.4, +0.3]	**+0.73** [+0.4, +1.1]	+0.18 [-0.2, +0.6]	**+0.78** [+0.2, +1.3]	**+1.27** [+0.3, +2.2]	+0.39 [-0.1, +0.9]	**+0.90** [+0.3, +1.4]
GeoK − facility location	+0.12 [-0.6, +0.8]	+0.29 [-0.3, +0.9]	**+0.70** [+0.1, +1.3]	-0.50 [-1.3, +0.3]	+0.55 [-0.1, +1.3]	+0.11 [-0.3, +0.5]	-0.37 [-1.0, +0.3]
GeoK − feature clusters	**-0.54** [-1.0, -0.0]	**+0.75** [+0.2, +1.4]	+0.46 [-0.0, +1.0]	+0.53 [-0.0, +1.1]	**+0.85** [+0.2, +1.5]	**+0.88** [+0.4, +1.4]	+0.32 [-0.5, +1.0]
GeoFL − facility location	n/a	n/a	n/a	+0.32 [-0.4, +1.0]	+0.45 [-0.4, +1.4]	+0.03 [-0.2, +0.3]	**+1.35** [+0.6, +2.1]
GeoFL − placebo weights	n/a	n/a	n/a	+0.93 [-0.2, +2.1]	+0.76 [-0.0, +1.6]	+0.12 [-0.2, +0.5]	**+2.13** [+1.3, +3.0]

cell: mean difference [95% paired region bootstrap] · outlined: interval excludes zero · n/a: weighted rules equal ordinary FL in balanced pools and were not fitted

b Frozen control: gains peak where the budget equals the number of source regions

